

Code Explanation

Configuration Module for E2E Policy Services YUYU Extract

The provided code is a configuration module for the E2E Policy Services YUYU extract pipeline. It manages database connections, parameters, and paths used in the pipeline.

Overview of the Code

The code defines a class named `Config` that encapsulates various configuration parameters and methods for constructing Oracle JDBC URLs, retrieving Oracle connection properties, and getting Spark configuration.

Step 1: Importing necessary modules and defining the Config class

The code starts by importing the necessary modules, including `os` for interacting with environment variables and `Dict` and `Any` from the `typing` module for type hinting. It then defines a class named `Config` to encapsulate the configuration parameters and methods.

```
import os
from typing import Dict, Any

class Config:
    """Configuration class for the YUYU extract pipeline."""
```

Step 2: Defining Oracle connection parameters

The `Config` class defines several class attributes to store Oracle connection parameters, such as host, port, service name, username, and password. These values are retrieved from environment variables with default values provided.

```
# Oracle connection parameters
ORACLE_HOST = os.getenv("ORACLE_HOST", "localhost")
ORACLE_PORT = os.getenv("ORACLE_PORT", "1521")
ORACLE_SERVICE = os.getenv("ORACLE_SERVICE", "ORCL")
ORACLE_USER = os.getenv("ORACLE_USER", "ZSYSE2EDEV")
ORACLE_PASSWORD = os.getenv("ORACLE_PASSWORD", "password")
```

Step 3: Defining processing parameters and output paths

The class defines additional attributes for processing parameters, such as `M_PROCESS_USERID`, and output paths, including `OUTPUT_BASE_PATH` and `FLAT_FILE_OUTPUT_PATH`.

```
# Processing parameters
M_PROCESS_USERID = os.getenv("M_PROCESS_USERID", "ETL_USER")

# Output paths
OUTPUT_BASE_PATH = os.getenv("OUTPUT_PATH", "/tmp/yuyu_output")
FLAT_FILE_OUTPUT_PATH = f"{OUTPUT_BASE_PATH}/DPAddressChangeYUYU"
```

Step 4: Defining table names and sequence generator settings

The class defines attributes for table names used in the pipeline, such as `SOURCE_TABLE`,

`LOOKUP\_TABLE\_YUYU\_CLNT`, and `LOOKUP\_TABLE\_YUYUK\_CLN`, as well as sequence generator settings.

```
# Table names
SOURCE_TABLE = "STG_E2E_AC_TXDBH_DATA"
LOOKUP_TABLE_YUYU_CLNT = "T_YUYU_CLNT"
LOOKUP_TABLE_YUYUK_CLN = "T_YUYUK_CLN"

# Sequence generator settings
SEQUENCE_START = 0
SEQUENCE_INCREMENT = 1
```

Step 5: Defining output format

The class defines an attribute `OUTPUT\_FORMAT` to store the output format, which is retrieved from an environment variable with a default value of "parquet".

```
# Output format
OUTPUT_FORMAT = os.getenv("OUTPUT_FORMAT", "parquet") # parquet or
CSV
```

Step 6: Implementing methods for constructing Oracle JDBC URL and retrieving connection properties

The class includes a class method `get\_oracle\_jdbc\_url` to construct the Oracle JDBC URL using the defined connection parameters.

```
@classmethod
def get_oracle_jdbc_url(cls) -> str:
    """Construct Oracle JDBC URL."""
    return f"jdbc:oracle:thin:@{cls.ORACLE_HOST}:{cls.ORACLE_PORT}/{
cls.ORACLE_SERVICE}"
```

Step 7: Implementing method for retrieving Oracle connection properties

The class includes another class method `get\_oracle\_properties` to retrieve Oracle connection properties, including username, password, and driver.

```
@classmethod
def get_oracle_properties(cls) -> Dict[str, str]:
    """Get Oracle connection properties."""
    return {
        "user": cls.ORACLE_USER,
        "password": cls.ORACLE_PASSWORD,
        "driver": "oracle.jdbc.driver.OracleDriver"
    }
```

Step 8: Implementing method for retrieving Spark configuration

The class includes a class method `get\_spark\_config` to retrieve Spark configuration properties.

```
@classmethod
def get_spark_config(cls) -> Dict[str, Any]:
    """Get Spark configuration."""
    return {
        "spark.sql.adaptive.enabled": "true",
        "spark.sql.adaptive.coalescePartitions.enabled": "true",
        "spark.sql.session.timeZone": "Asia/Tokyo"
    }
```